

Tony Antoine Abdo

tonyabdo.com

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Education

University of California, Berkeley

M.S. - Mechanical Engineering | *Concentration in Controls*

May 2017

B.S. - Mechanical Engineering | *Minor in Electrical Engineering and Computer Science*

May 2016

Relevant Coursework: Feedback Controls, Model Predictive Control, Mechanical Behavior of Engineering Materials, Artificial Intelligence, Machine Learning, Robotics, Embedded Systems, Vehicle Dynamics, Signals and Systems

Relevant Experience

Robotics System Developer – AlphaZ (Start-up)

Feb 2025 – Present

- Implement Python and CPP ROS1 packages to automatically traverse uncertain terrain with obstacle avoidance demonstrating capability on a 1.5 mile loop after rainfall with 0 intervention on customer's property
- Design, build, and deploy various payloads on legged and wheeled robots for security purposes
- Develop auto-docking sequence achieving 100/100 successful docking attempts for quadruped robots using vision and closed loop control strategies with sequencing controlled via a state machine
- Debugged critical problems in ICP and planning algorithms that had gone unnoticed by other team members

Co-Founder – Friends of Falkon Robotics Inc. 501 (c)(3)

Feb 2021 – Dec 2025

- Created a non-profit with a group of alumni to facilitate greater STEM education for High School students
- Search for and complete grants to fund team with a budget that has grown from ~\$30k to \$100k+

Co-Founder – Leaf Suit Inc. YC W20 (Wearable Air Conditioner)

Jun 2018 – Jan 2021

- Designed, built, and programmed a 1" x 2" PCBA controlling 6 fans, 2 diaphragm pumps, and smart button
- Designed and made pattern to allow comfort for wearer while still achieving conformity and cooling coverage
- Worked with professional grant writers to define targets and put together grant submissions for NHS and military
- Worked with patent attorney to draft a non provisional patent along with PCT filing

Engineering Mentor (FRC/Berkeley PiE)

Spring 2014, Jan.– Mar 2018, Dec 2021 – Dec 2025

- Teach students how to design and build a robot while taking into account real physical principles
- Develop design and manufacturing workflow to allow a 100+ student organization to create a robot effectively

Mechatronics Engineer – Tesla

Apr 2018 – Dec 2019

- Conceptualized, designed, and brought up manufacturing of plastic parts in over 50,000 Model S/X vehicles
- Designed and built machines for operators to use in seats manufacturing, increasing production by 30%
- Developed a controller utilizing CAN protocol to command seat motions when isolated from the car

Hardware Engineer – Dash Robotics Internship

Jun 2016 – Aug 2016

- Self taught C to prototype, design, and implement the firmware infrastructure needed to support future accessories on the Kamigami robot's existing i2c communication bus to boost future sales
- Designed and manufactured tools to aid in improving the assembly line yield from 70% to 90%

Technical Skills

- 3D Modeling: Autodesk Inventor, CATIA, SolidWorks FEA, SolidWorks certified
- Programming: Python, C++, ROS1, Linux (proficient) | MATLAB, ROS2 (limited)
- Other: Altium PCB design, machining on lathe and mill, 3D printing, CNC Router, soldering, i2c, SPI, CAN, LIN

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Projects

Line Following Car

- Designed and manufactured circuit boards in Eagle for BLDC motor control and power regulation
- Wrote the firmware for reading sensor data within an interrupt-driven control loop for accurate and tight timing
- Incorporated embedded feedback controls with gain scheduling for steering angle and velocity control

Drone Controller

- Derived 3D physics model of a quadcopter using matrix representation of the dynamics
- Implemented various control and state estimation techniques determined from MATLAB simulations

Adaptive Cruise Control (ACC)

- Designed the electrical system for two cars to drive autonomously with off-the-shelf components
- Implemented the embedded controls for both the lead and follow cars, as well as the data collection infrastructure